Minimum central circles: an effective characterization of the global asymptotic constant

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Abstract

Let $R^*(n)$ be the minimum radius of a central circle externally tangent to nonoverlapping circles of radii $1, \dots, n$. We prove that $R^*(n) = C_* n^2 + o(n^2)$ and give finite linear programs that enclose C_* with prescribed error. An exact angular comparison reduces the problem to marked points on a line with separation at least the geometric mean of each pair of marks. Concatenation with bounded joining cost and reassignment to genuine labels prove existence of the limit. Balanced finite-word programs give a bracket of width $(1/k+1/r)/\pi$, before directed arithmetic error, using k types and words of length r. For explicit geometric upper bounds, we recover finite and countable reflected-block constructions by actual permutations, retaining all pair constraints. A coupled terminal-subset argument gives an explicit lower bound strictly above $1/8$. Together these give $C_{\mathrm{term}} + \eta_{\mathrm{width}} \le C_* \le U_4$, with neither endpoint asserted sharp. The finite-word scheme establishes computability; efficient high-precision evaluation and the structure of global optimizers remain open.

Keywords: circle packing; geometric optimization; asymptotic limits; linear programming; rigorous computation.
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1 Problem and Relation to Prior Finite Work

All outer circles are tangent to the central circle; outer circles need not be tangent to each other. For a cyclic order σ, the adjacent chain radius $R_{\mathrm{chain}}(\sigma)$ ignores nonadjacent constraints, whereas $R_{\mathrm{full}}(\sigma)$ is the least radius satisfying every pair constraint in that order. The global quantity is $R^*(n) = \min_\sigma R_{\mathrm{full}}(\sigma)$. Always $R_{\mathrm{chain}}(\sigma) \le R_{\mathrm{full}}(\sigma)$. A feasible construction supplies an upper bound, not an optimum certificate.

Paper I, in its corrective version [1], develops the angular model, the anti-Monge/Supnick chain-ordering connection and finite optimization. Its reported finite certification range is $3 \le n \le 14$; these finite computations are not premises of the present asymptotic proofs. The original version [2] proposed the coefficient $1/8$, which the corrective v2 identifies as disproved by the sequel. We cite v2 for the current finite-paper account and retain v1 only for this historical claim. No finite table or global floating-circle classification is reproduced here.

The scientific results here appeared in the standalone preprint arXiv:2609.13630 [3]. This separate journal manuscript supplies autonomous proof supplements and makes their dependencies explicit; it does not replace either public preprint. The current package's contribution over that preprint is proof exposition and reproducibility, not a new asymptotic coefficient.

Our main result is an effective characterization of the true global leading coefficient. Sections 2–4 give its proof directly from the angular model, through an all-pair line problem, genuine-label concatenation and balanced finite-word programs. The lower and upper bounds converge to the same constant with a prescribed error. Separate explicit constructions give the interval stated in the abstract and disprove the coefficient conjecture of [2]. Their complete proof dependencies are given in the supplementary material; the main variational characterization does not assume either endpoint mechanism is optimal.

The line problem has a familiar geometric interpretation. Disks of radii a_i tangent to the horizontal axis, with centers $(2x_i, a_i)$, are nonoverlapping precisely when $|x_i - x_j| \geq \sqrt{a_i a_j}$. For $a_i \leq 1$, their horizontal envelope span differs from $2(\max_i x_i - \min_i x_i)$ by at most two. Alt et al. [4] study minimum-width shelf packing and give approximation, special-case exact algorithms and hardness results. The shelf model and its geometric-mean separation rule are pre-existing. Our claim concerns the uniform-label asymptotic regime and its transfer to central-circle geometry, with a balanced-word error bound. General shelf-packing hardness does not imply hardness of evaluating this particular constant.

Another related setting is weighted disk-contact recognition. Klemz et al. [5] study prescribed disk radii and contact graphs, including stars with arbitrary weights. Ringmin instead fixes the outer radii to $1, \ldots, n$, varies the central radius, and permits additional outer contacts. We make no transfer of their recognition hardness to evaluation of C_*.

Likewise, difference-constraint feasibility is classical [6]; its use for a fixed Ringmin order is already part of Paper I. We do not claim a new general method for such systems. The contribution of the reflected-block theorem is a special all-pairs reduction and quantitative recovery by genuine labels. The comparison here is scoped to these directly related sources, not an exhaustive priority survey.

2 Exact Angular Comparison

The cosine law gives the required pair angle

$$\theta_R(u, v) = 2 \arcsin \sqrt{\frac{uv}{(R+u)(R+v)}} = 2 \arctan \sqrt{\frac{uv}{R(R+u+v)}}. \tag{1}$$

Both directed cyclic arcs between a pair must be at least this angle. This is equivalent to nonoverlap, since their smaller separation belongs to $[0, \pi]$. The angle is positive, symmetric, increasing in either outer radius and strictly decreasing in $R > 0$. For a cyclic order $\sigma = (u_1, \ldots, u_n)$, $n \geq 3$, with $u_{n+1} = u_1$, $R_{\text{chain}}(\sigma)$ is the unique positive root of $\sum_{i=1}^{n} \theta_R(u_i, u_{i+1}) = 2\pi$. Indeed, the sum decreases continuously from $n\pi$ to zero. Summing the adjacent constraints in any full placement proves $R_{\text{chain}}(\sigma) \leq R_{\text{full}}(\sigma)$.

Definition 1. For a finite multiset $a = (a_1, \ldots, a_N)$ of marks in $[0, 1]$, let $b(a)$ be the least span of a line placement satisfying

$$|x_i - x_j| \geq \sqrt{a_i a_j} \quad (i \neq j). \tag{2}$$

Distinct zero-mark points may coincide. Write $b_n = b(1/n, 2/n, \ldots, 1)$.

For a fixed word $w = (w_1, \ldots, w_N)$ in left-to-right order, its minimum span $\ell(w)$ is attained by

$$x_1 = 0, \qquad x_i = \max_{j<i} \{x_j + \sqrt{w_i w_j}\}, \qquad \ell(w) = x_N. \tag{3}$$

Induction proves that these coordinates are componentwise no larger than any feasible ordered coordinates with $x_1 = 0$; they also satisfy every constraint. Nonnegative weights ensure ordered coordinates. Equivalently, $\ell(w)$ is the longest increasing-index path from 1 to N, with edge weight $\sqrt{w_i w_j}$. Thus $b(a)$ is an attained minimum over a finite set of orders. Deleting or decreasing marks cannot increase it; scaling all marks by h scales the minimum span by h.

Lemma 2 (Bounded closing cost)**.** *Let $B(a)$ be the infimum circumference with both pair arcs at least $\sqrt{a_i a_j}$. Then $b(a) \le B(a) \le b(a) + 1$.*

Proof. Cut a feasible circle to obtain a feasible line placement. Conversely, close a feasible line placement by an additional gap of length one. One arc for every pair is its line separation; the other contains the added gap and is at least $1 \ge \sqrt{a_i a_j}$. Both arcs are controlled. □

Lemma 3 (Exact geometric squeeze)**.** *For every $n \ge 3$,*

$$\frac{b_n}{\pi n} - \frac{1}{n} \le \frac{R^*(n)}{n^2} \le \frac{b_n + 1}{\pi n}. \tag{4}$$

Proof. For $1 \le u, v \le n$, the two expressions in (1) give

$$\frac{2\sqrt{uv}}{R+n} \le \theta_R(u,v) \le \frac{2\sqrt{uv}}{R}.$$

The first uses $\arcsin z \ge z$ and $\sqrt{(R+u)(R+v)} \le R+n$; the second uses $\arctan z \le z$. Scaling a feasible ring's angles by $(R+n)/(2n)$ gives a feasible linear circle of circumference $\pi(R+n)/n$, proving the lower inequality. Conversely, the circumference $b_n + 1$ from Lemma 2, scaled to 2π, satisfies every original pair angle at $R = n(b_n+1)/\pi$. This proves the upper inequality. No Taylor expansion, adjacency reduction or assumption on the optimal order was used. □

3 Existence by Genuine-Label Concatenation

Theorem 4 (The leading limit exists)**.** *There is a constant C_* such that*

$$R^*(n) = C_* n^2 + o(n^2), \qquad \pi C_* = E := \lim_{n\to\infty} \frac{b_n}{n} = \inf_{n \ge 2} \frac{b_n + 1}{n}. \tag{5}$$

Proof. Copies of a feasible line template, separated by unit gaps, satisfy all cross-copy constraints as well as the internal ones. Fix $n \ge 2$. Given $N \ge n$, let $q = \lceil N/n \rceil$ and concatenate q copies of an optimal b_n template. Its span is $q(b_n + 1) - 1$. Retain the N smallest marks. The ith sorted retained mark is $\lceil i/q \rceil / n$. Scale all marks and coordinates by $h = nq/N$. The resulting ith mark is at least i/N, so decreasing it to i/N preserves every constraint. Each target label is assigned exactly once to an actual retained point, with ties broken arbitrarily. Hence

$$b_N \le \frac{nq}{N}\big[q(b_n + 1) - 1\big]. \tag{6}$$

Intermediate marks exceeding one cause no problem: the line inequalities are homogeneous, and final marks are exactly $1/N, \ldots, 1$.

As $N \to \infty$, $q/N \to 1/n$, giving $\limsup_N b_N/N \le (b_n + 1)/n$ for every fixed n. The sequence is bounded between zero and one by equally spaced placements. Taking n along a subsequence attaining its liminf gives equality of limsup and liminf. Taking the infimum and then $n \to \infty$ proves the last equality in (5). Lemma 3 transfers the limit to the original ring problem for all sizes and both parities. Finally, at least $n/2$ marks are at least $1/2$. Their successive line separations are at least $1/2$, so $E \ge 1/4$ and $C_* > 0$. □

4 Effective Finite-Word Characterization

4.1 Finite Types and a Uniform Quantization Error

For $k \geq 1$, let $A_{k,q}$ contain q copies of each mark $1/k, \ldots, 1$, and let $A^-_{k,q}$ instead contain q copies of $0, 1/k, \ldots, (k-1)/k$. Define $e_k = \lim_{q\to\infty} b(A_{k,q})/(kq)$ and $e^-_k = \lim_{q\to\infty} b(A^-_{k,q})/(kq)$; these limits exist. Indeed $a_q = b(A_{k,q}) + 1$ is subadditive by unit-gap concatenation. Writing $q = hp + s$ at fixed p proves $\limsup a_q/q \leq a_p/p$; then taking p along a liminf subsequence proves the limit. The same argument applies to the lower types.

The two multisets share every nonzero type except the top type. Zero marks cost nothing. Adding the q unit marks as a separate equally spaced block costs at most q, including the join. Thus, also for $k = 1$,

$$e^-_k \leq e_k \leq e^-_k + \frac{1}{k}.$$

For $n = kq$, the sorted uniform marks i/n lie between the two sorted type multisets. Theorem 4 and monotonicity imply

$$e_k - \frac{1}{k} \leq E \leq e_k. \tag{7}$$

4.2 Balanced Words and the Quantified Boundary Relaxation

Fix $k \geq 1$ and $r \geq 2$, and use all k^r words $w \in \{1/k, \ldots, 1\}^r$, including repetitions. Let $c_j(w)$ count occurrences of j/k. Compute $\ell(w)$ by (3), and define

$$\begin{aligned} \lambda_{k,r} = \min_p \ & \frac{1}{r} \sum_w p_w \ell(w), \\ p_w \geq 0, \quad & \sum_w p_w = 1, \quad \sum_w p_w c_j(w) = \frac{r}{k} \quad (1 \leq j \leq k). \end{aligned} \tag{8}$$

The polytope is nonempty: equally mix the k constant words.

Theorem 5 (Matching effective variational bounds)**.** *For every $k \geq 1, r \geq 2$,*

$$\boxed{\frac{\lambda_{k,r} - 1/k}{\pi} \leq C_* \leq \frac{\lambda_{k,r} + 1/r}{\pi}}. \tag{9}$$

Consequently $C_* = \pi^{-1} \lim_{k,r\to\infty} \lambda_{k,r}$, jointly, with certified bracket width $(1/k + 1/r)/\pi$.

Proof. Partition an optimal ordering of $A_{k,q}$ into consecutive blocks of r vertices, discarding fewer than r leftover vertices. Their spans sum to at most the full span and each is at least its word cost. Empirical word distributions have convergent subsequences in the finite simplex. Their limiting mean type counts are r/k, since the discarded counts vanish after normalization. Hence $\lambda_{k,r} \leq e_k$.

Conversely, the rational feasible polytope in (8) has a minimizing rational vertex even though its cost entries are algebraic. Choose a common denominator D. A batch with Dp_w copies of each word has exactly Dr/k copies of each type, an integer by the count constraints. Place every word at its minimum span and separate blocks by unit gaps. All within-block and cross-block pairs are feasible. Repeating batches gives asymptotic mean span $\sum_w p_w(\ell(w) + 1)/r = \lambda_{k,r} + 1/r$. The finite-type limit therefore satisfies $e_k \leq \lambda_{k,r} + 1/r$. Combine with (7). □

4.3 Exact Certificates and Computational Meaning

Directed rational square-root enclosures of width ϵ give word-span enclosures of width at most $(r-1)\epsilon$: every path in (3) has at most $r-1$ edges. Rational endpoint LPs then enclose $\lambda_{k,r}$ with additional width at most $(r-1)\epsilon/r$. They can be solved by finite exact vertex enumeration. Practical solvers may instead find solutions for subsequent exact verification. For a rational lower cost $\ell^-(w)$, dual coefficients satisfying

$$y_0 + \sum_j y_j c_j(w) \le \ell^-(w)/r \quad \text{for every word}$$

certify the lower value $y_0 + (r/k)\sum_j y_j$. Feasible rational p certifies an upper value using upper costs. All such checks are finite rational comparisons; equality is not decided by indefinite interval refinement. Together with a rational enclosure of π, this proves arbitrary-precision computability of C_* in principle.

The word count is exponential in r. No efficient high-precision algorithm or closed-form constant follows. The supplementary implementation uses SciPy to find sparse primal and dual solutions; exact rational arithmetic validates every equality and word inequality. Small word programs need not improve the stronger explicit endpoint bounds below.

5 Explicit Geometric Upper Constructions

Fix $0 \le \alpha < 1$ and nonnegative adjacent slab lengths $\ell_1, \dots, \ell_k$ of total $T < 1-\alpha$. Put $A = 1+\alpha$, $t_0 = 0$, $t_j = \sum_{h\le j} \ell_h$. On each slab $[t_{j-1}, t_j]$, use equal masses at

$$(t, A+t, A+t_{j-1}+t_j-t), \qquad (t, A+t_{j-1}+t_j-t, A+t).$$

After T, use the diagonal $(t, 1+\{t+\alpha\}, 1+\{t+\alpha\})$. This defines a measure μ on $[0,1]\times[1,2]^2$ with uniform low and separate high marginals. Its complete cell cost is

$$C(\mu) = \frac{1}{4\pi}\int \max\{\sqrt{t}(\sqrt{X}+\sqrt{Y}), \sqrt{XY}\}\, d\mu. \tag{10}$$

Correct marginals alone do not establish finite recovery.

Theorem 6 (General block transfer)**.** *Every such finite partition has genuine alternating-half permutations with all-pairs-feasible radii $\rho_m = R_{\mathrm{full}}(\sigma_m)$ and*

$$\left|\frac{\rho_m}{(2m)^2} - C(\mu)\right| \le \frac{(6k^2+28k+1060)/m + 16384/(3m^2)}{4\pi}$$

for $m \ge \max\{2048, \lceil 2/(1-\alpha-T)\rceil\}$. It follows that $C_* \le C(\mu)$. Countably many slabs also transfer if their total length is strictly below $1-\alpha$.

Proof with explicit construction. Let $s = \lfloor \alpha m \rfloor$, $l_j = 2\lfloor \ell_j m/2 \rfloor$, $a_0 = 0$ and $a_j = \sum_{h\le j} l_h$. Reverse only the even ranks within each $(a_{j-1}, a_j]$ to obtain a permutation J of $\{1,\dots,m\}$. Set $J(i) = i$ on odd ranks and on every index outside these blocks. Set $P_i = m+1+((J(i)+s-1) \bmod m)$ and $\sigma_m = (1, P_1, \dots, m, P_m)$, with $P_0 = P_m$. Disjoint involutions and the cyclic shift prove bijectivity, including zero blocks and length-two identities. The lower bound on m keeps the final block endpoint at least two ranks before the high wrap, and its exit strictly before that wrap.

Double panels of width $2/m$, allocated half to each parity, recover the two reflected orientations. Interior coordinate error is at most $4/m$. There are at most $k+3$ exceptional actual cells: shared exits, the cyclic predecessor, and the two wrap cells. Rounded slab boundaries and wrap differ on a set of length at most $[k(k+1)+1]/m$, with remaining coordinate error at most $(4k-1)/m$. Write $g(t,X,Y) = \max\{\sqrt{t}(\sqrt{X}+\sqrt{Y}), \sqrt{XY}\}$ for the integrand in (10). It is 4-Lipschitz and lies in $[1,3)$. The actual normalized full-cell moment is

$$G_m = \frac{1}{m}\sum_{i=1}^{m} g(i/m, P_{i-1}/m, P_i/m).$$

It differs from $\int g\,d\mu$ by at most $(6k^2+28k+36)/m$.

For arbitrary distinct highs in $[m+1, 2m]$, the exact full-feasibility criterion for this alternating order is

$$\sum_{i=1}^{m} \max\{\theta_R(P_{i-1}, i) + \theta_R(i, P_i), \theta_R(P_{i-1}, P_i)\} \le 2\pi.$$

Its sufficiency follows from the high-shell triangle inequality $2\theta_R(m+1, m+1) > \theta_R(2m, 2m)$ and high/low separation: each directed high path contracts through whole cells, and paths with low endpoints either contain such a high path or an incident high gap that already exceeds the endpoint requirement. Both directions are treated separately. At its unique root, choose the first half of each cell at its required low/high angle and put any excess in the other half. This closes at 2π and satisfies all pairs.

At $R = 4cm^2$, $c \ge 1/32$, the full score differs from $G_m/(2c)$ by at most $1024/m + 16384/(3m^2)$. For $m \ge 2048$ this first brackets the actual normalized root in $(1/32, 1/2)$, then yields the stated error. Deleting label $2m$ gives the odd upper bound. For countably many slabs, truncate and use that the omitted complete-cost integral is at most six times the omitted length. A diagonal sequence with m growing faster than k^3 pays both finite and tail errors. Full cell/seam formulas and the angle estimate are supplied in the versioned standalone upper supplement S2 [8]. □

For the explicit four-block construction, let x_* be the unique minimizer on $[0,1]$ of

$$E(x) = \int_0^x \left[\max\{\sqrt{(1+t)(1+x-t)}, \sqrt{t}(\sqrt{1+t}+\sqrt{1+x-t})\} - \max\{1+t, 2\sqrt{t(1+t)}\}\right] dt.$$

Let $K(\alpha) = \int_0^1 \max\{\sqrt{t h_\alpha(t)}, h_\alpha(t)/2\}\,dt$, where $h_\alpha(t) = 1 + \{t+\alpha\}$, and let $\widehat{\alpha}$ be the unique zero of $K'(\alpha) + (1+\alpha)E(x_*)$ on $[0, 1/2]$. Set $A = 1+\widehat{\alpha}$ and $\lambda = Ax_*$. The second width ϵ_b is the unique mixed-branch stationary minimum for a separate reflection starting at λ; the third width Δ_* is the analogous unique minimum starting at $v = \lambda + \epsilon_b$. Each width is minimized over $[0, A/3 - u]$ at its fixed start u, retaining the full max. The supplement [8] gives their exact defining integrals, existence/uniqueness proofs and rational isolations:

$$.1093 < \widehat{\alpha} < .10931, \qquad .2876 < x_* < .2877,$$
$$.043 < \epsilon_b < 7/160, \qquad 29/5000 < \Delta_* < 27/4000.$$

These terminating decimals are rational bracket endpoints, not definitions or numerical replacements for the parameters.

Let $C_3(d)$ be (10) for the three lengths (λ, ϵ_b, d), and $C_4(\eta)$ the same cost for the four lengths $(\lambda, \epsilon_b, \Delta_*, \eta)$. Use $\eta = 1/20000$ and define $U_4 = C_4(1/20000)$. If $w = \lambda + \epsilon_b + \Delta_*$, $B = A + w$ and $M = B + \eta/2$, the fourth increment at $0 < \eta \le 1/20000$ is exactly

$$C_4(\eta) - C_3(\Delta_*) = -\frac{1}{4\pi}\int_{-\eta/2}^{\eta/2} \frac{z^2}{M + \sqrt{M^2 - z^2}}\,dz.$$

Rational input margins keep both changed maxima strictly chord. Since $2B \le M + \sqrt{M^2 - z^2} \le 2M$, $M < 3/2$ and $\pi < 4$,

$$C_* \le U_4 < C_3(\Delta_*) - \frac{1}{4608000000000000}. \tag{11}$$

The leading decrement is $-\eta^3/(96\pi B)$, with remainder between zero and $\eta^4/(192\pi B^2)$. The theorem transfers this literal continuous saving to genuine full geometry. It does not make four blocks globally optimal or optimize an infinite partition.

6 An Explicit Global Lower Endpoint

Define τ as the unique root of $\cos\tau = \tau$ in $(0,1)$, set $q = (1-\sin\tau)/(1+\sin\tau)$ and $C_{\mathrm{term}} = \tau(1+q)/(2\pi)$. For the fixed cutoffs

$$(\beta_1, \beta_2, \beta_3) = (2091/10000, 10907/50000, 23/100)$$

put

$$D_i = \int_q^{\beta_i} [1 + q - x - 2\sqrt{x(1+q-x)}]\, dx, \quad F(h) = \sum_i h_i D_i - 8\sum_i h_i^3.$$

Let $a = 113/12500$, $b = 593/50000$. On $h_i \ge 0$, $h_1 + h_2 \le a$, $h_2 + h_3 \le b$, define

$$\eta_{\mathrm{width}} = \max_h \frac{\max\{F(h), 0\}}{\pi(16 + 432\sum_i h_i)}.$$

This maximum is uniquely attained at $(a - z_*, z_*, b - z_*)$, where z_* is the unique zero on $[0, a]$ of $G = N'L + 432N$, $N(z) = F(a-z, z, b-z)$ and $L(z) = 16 + 432(a + b - z)$.

Proposition 7 (Explicit global lower bound)**.** *The global constant satisfies $C_* \ge C_{\mathrm{term}} + \eta_{\mathrm{width}}$, with exact directed enclosure*

$$.14056946887766098063257 < C_{\mathrm{term}} + \eta_{\mathrm{width}} < .14056946887766098063392.$$

In particular $C_* > 1/8$, so both the v1 coefficient conjecture and its stronger $n^2/8 - R^*(n) = O(\sqrt{n})$ claim are false.

Proof outline. The complete proof L1–L8 and exact arithmetic certificates are in the unchanged standalone lower supplement S1 [7]. Its essential coupled information is one shared crossing energy for the same order and all three nested terminal restrictions. An explicit assignment dual bounds the reflected-grid energy by eight times the excess chain cost. The absolute signed deletion discrepancy at cutoff i is at most 54 times this energy plus twice its crossing term. Separated strips charge the common energy once, yielding the scalar gain $\max\{F(h), 0\}/(16 + 432\sum h_i)$. The exact angular squeeze transfers each induced chain bound to any original full feasible configuration. Uniform finite errors vanish at every fixed strictly separated width vector.

For the width optimum, $G' < 0$ and rational endpoint bounds isolate its zero. Set $u = (a - z_*, z_*, b - z_*)$, $\mathcal{L}(h) = 16 + 432\sum_i h_i$ and $\rho = F(u)/\mathcal{L}(u)$. The gradient of $F - \rho\mathcal{L}$ at u is a positive combination of the two constraint normals. Its exact remainder is $-8\sum_i (h_i - u_i)^2(h_i + 2u_i)$, proving global uniqueness on the whole closed region. To pass to the geometric limit, first use tu at fixed $0 < t < 1$, take $n \to \infty$, then $t \uparrow 1$. The boundary widths fail finite floor separation infinitely often; the proof never uses them as an eventual finite separation condition. The uniform finite error bounds for this limiting argument are supplied in [7]. The result does not assert that this particular lower mechanism reaches C_*. □

7 Reproducibility and Open Questions

The review package contains this manuscript, the unchanged Proposition 7 supplement S1 [7], the standalone upper supplement S2 [8], and a claim–lemma–hypothesis–proof–verification map. The limit and finite-word proofs are complete in Sections 3–4; S1 independently includes limit existence for its own use. S2 includes the implicit-parameter definitions, their isolation gates, all-pairs feasibility and geometric transfer. No task-log reconstruction is a proof prerequisite.

The package includes isolated build instructions, source hashes and bounded checks for line placements, label reassignment and rational word-LP certificates. SciPy is used only for LP candidate discovery; exact rational checks validate the proposed primal and dual data. S1 and the upper endpoint sign gates use Python's standard library. Negative controls test deliberately omitted pair constraints, missing closing gaps and corrupt certificates. These finite checks corroborate their declared arithmetic scopes; the written proofs establish universal quantifiers and limiting statements. The source manifest identifies the precise manuscript revision and the reused supplement baseline. No external acceptance is asserted.

The leading normalized behavior is characterized effectively, while neither explicit endpoint is known to equal C_*. Practical evaluation or simplification of the constant, sharp explicit constructions, microscopic optimizer structure and subleading terms remain open. The number of words in the finite programs is exponential in their length; computability does not give an efficient high-precision algorithm. The limit theorem neither certifies a finite optimum beyond $n = 14$ nor establishes a universal floating-circle cascade or a contact structure for global optimizers.

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